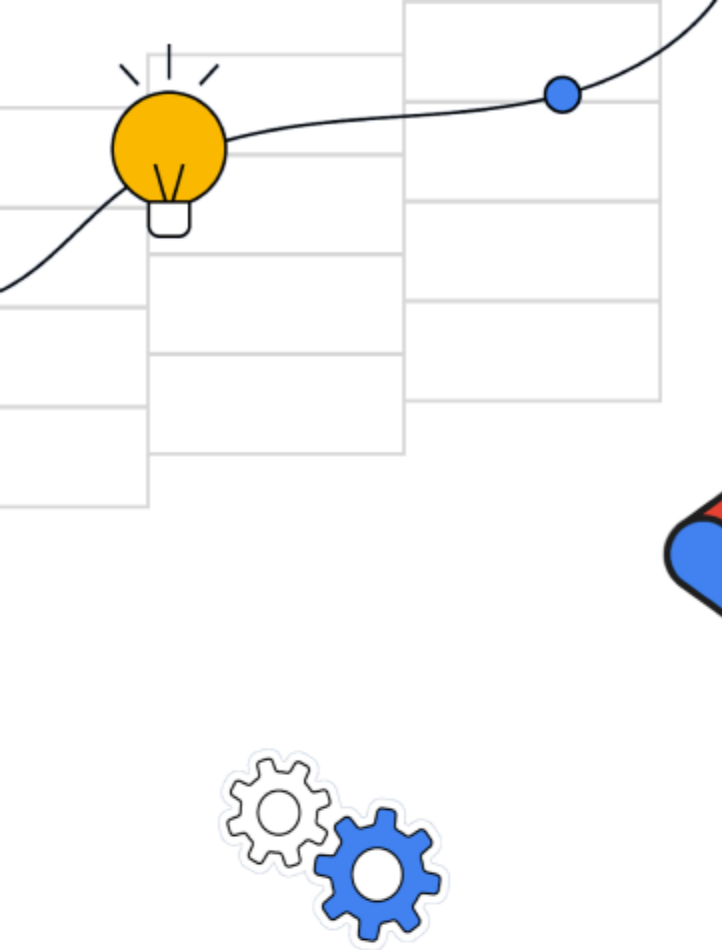
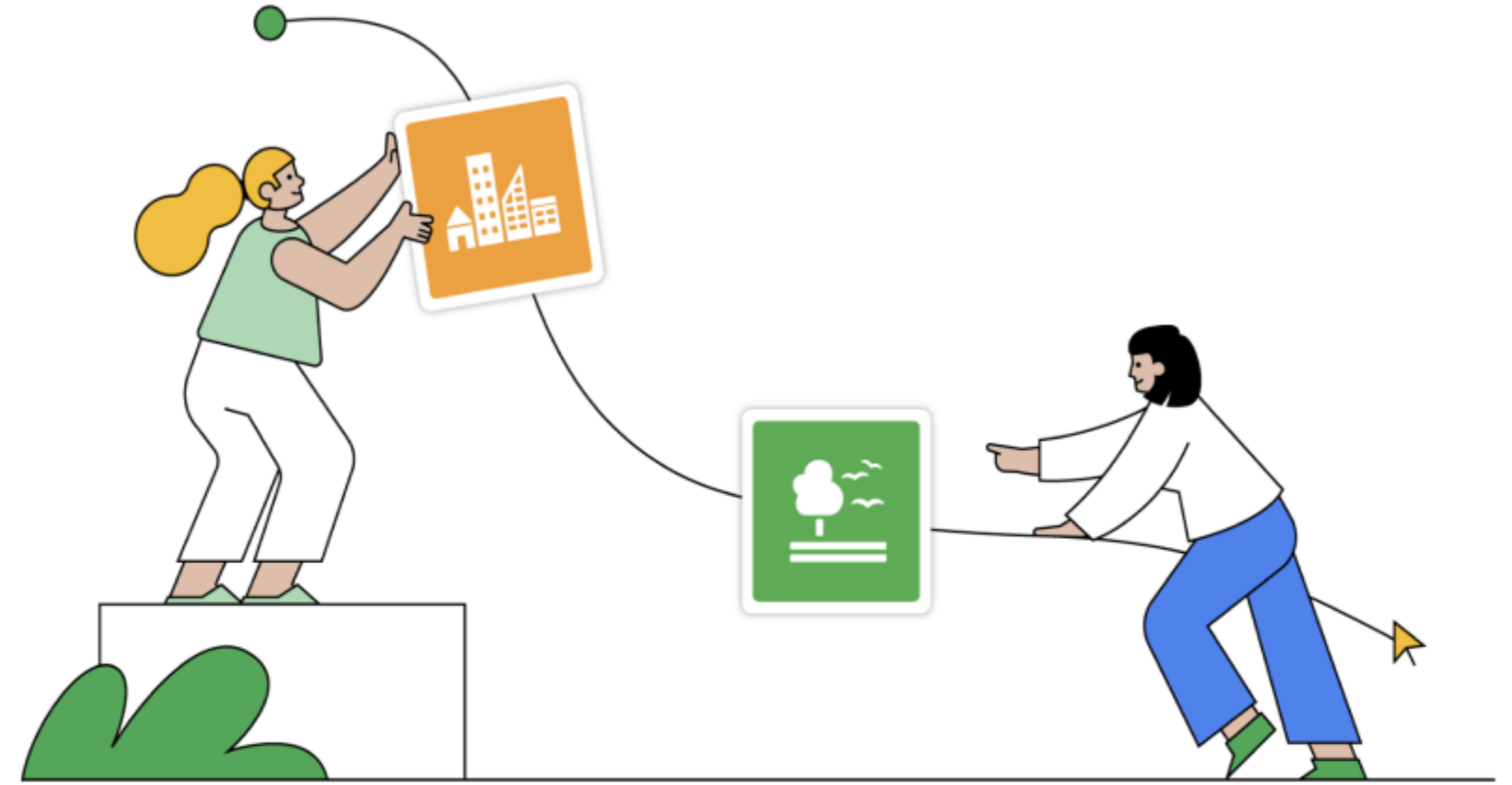




Solution Challenge



Team Details

- a. Team name: Team_CrowdInfra
- b. Team leader name: Tanishq Aswar
- c. Problem Statement: Open Innovation

Description:

- **Community-Driven Insight:** Users pin and upvote service needs (hospitals, ATMs, etc.)
- **Crowdsourced Demand:** Visualize where services are most needed.
- **AI-Powered Analysis:** Gemini API evaluates competition, profitability & resources.
- **Property Marketplace:** Links property owners with expanding businesses.
- **Tech Stack:** Next.js, Node.js/Express.js, MongoDB.
- **User-Friendly Visuals:** Google Maps API shows demand clusters & gaps.

How It's Different:

- **Live Crowd Data:** Real-time input beats outdated surveys.
- **AI Insights:** Gemini API powers deep, real-time analysis.
- **Dual Purpose:** Combines demand mapping + property marketplace.

Problem Solving:

- **Targets Real Gaps:** Uses actual demand to spot infrastructure needs.
- **Smart Decisions:** Data-backed insights on market & resources.
- **Boosts Expansion:** Connects businesses with ideal properties.

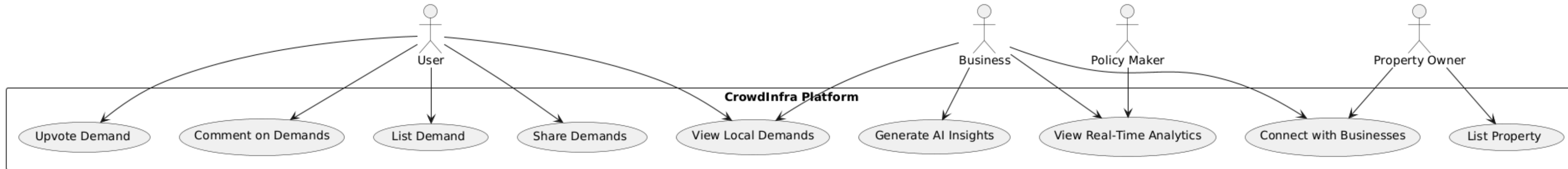
USP:

- **All-in-One:** Demand mapping, AI analysis, and listings in one place.
- **Dynamic & Scalable:** Real-time, precise, and future-ready.

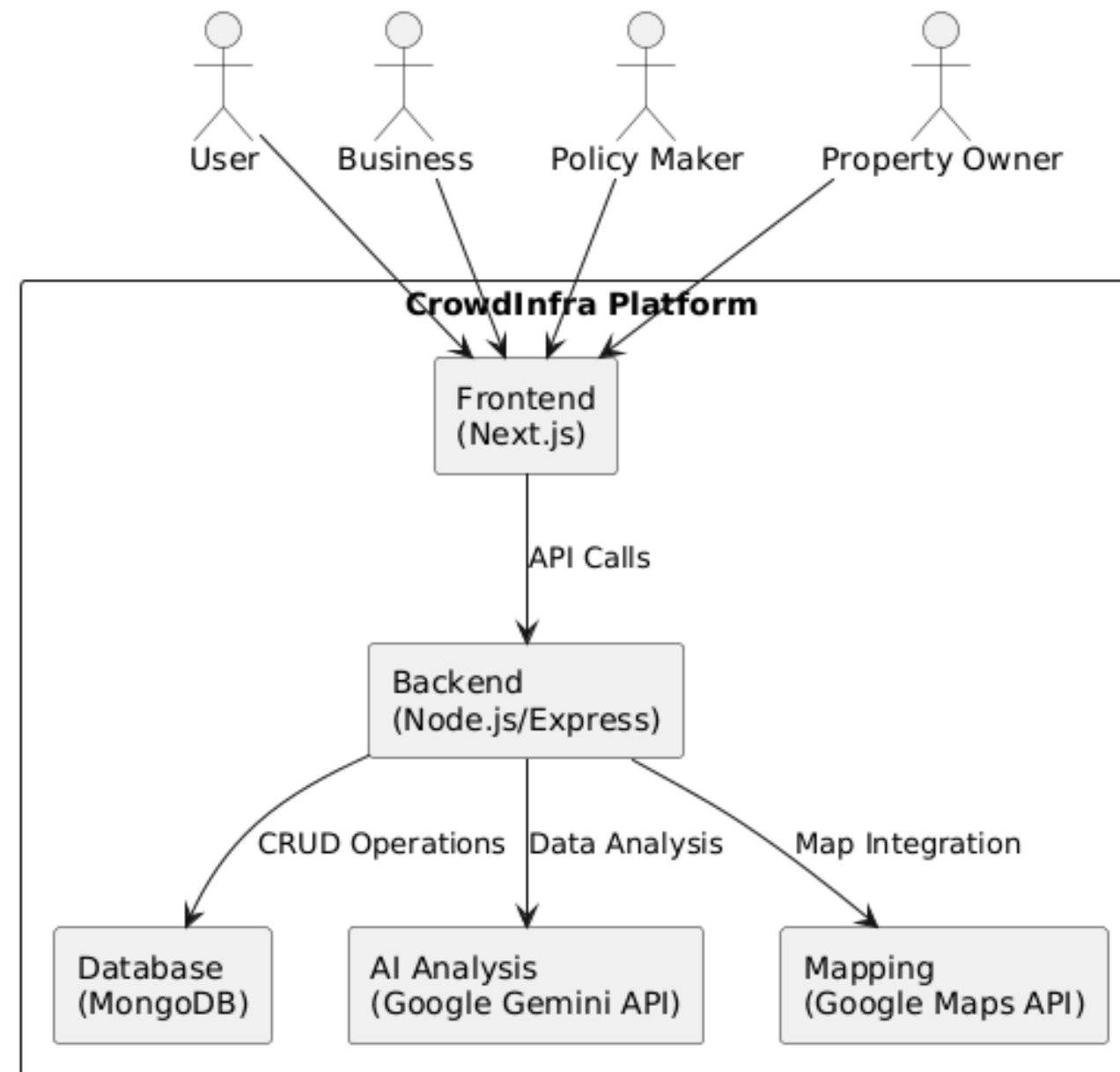
Key Features:

- **Pin Locations:** Mark where services are needed.
- **Upvote Demands:** Community ranks priorities.
- **AI Analytics:** Gemini API analyzes competition & feasibility.
- **Property Marketplace:** Connects landowners with businesses.
- **Interactive Map:** Google Maps shows demand clusters & gaps.
- **User Feedback:** Comments reveal the “why” behind demands.
- **Contextual Insight:** Feedback adds depth & credibility to trends.

Use-case diagram:



Architecture diagram:



Technologies used in the solution:

Frontend:

- Next.js: Delivers a blazing-fast, SEO-friendly, interactive user interface.

Backend:

- Node.js & Express: Efficiently handles API requests and website logic.

Database:

- MongoDB: Scalable and flexible for storing community demand data, property listings, and more.

Mapping:

- Google Maps API: Visualizes demand hotspots and infrastructure gaps with interactive maps.

AI Analysis:

- Google Gemini API: Provides deep, real-time insights on market competition, profitability, and resource availability.

Snapshots of the MVP:

<https://drive.google.com/drive/folders/1U-NyO2vEY45Ps-9mBNcHgH2Rj7egfcXX?usp=sharing>

Additional Details/Future Development:

- **Radius Selection:** Map demand over broader areas, not just points.
- **Smarter AI:** Optimize Gemini API for faster, sharper insights.
- **Trusted Data:** Integrate reliable datasets for accurate analysis.
- **UI/UX Boost:** Smoother, more intuitive user experience.
- **Scale & Secure:** Build and grow with security.
- **Local Suggestions:** Recommend businesses based on real demand.
- **Smart Property Matches:** Suggest listings in high-demand zones.
- **Data Validation:** Ongoing spam protection for quality input.

Project Links:

GitHub Public Repository: <https://github.com/TanishqAswar/CrowdInfra>

Demo Video Link: <https://www.youtube.com/watch?v=oVtXzXR5p3w>

MVP Link: <https://infra-crowd.vercel.app/>



Thank you

